

Numerical PDEs Homework #0
Due Tuesday, September 15, 2026

Problem 1: Finite differences Write a matlab code that uses finite differences to compute the first derivative of

$$f(x) = \exp(\sin(x))$$

on the interval $0 \leq x \leq 2\pi$. Use second-order centered differences on internal grid points and first-order one-sided differences at endpoints.

- (a) Show a graph of the exact analytic solution $f'(x)$ and your numerical solution $f'_n(x)$.
- (b) Show a relative error convergence plot by plotting the relative error on the y-axis and the grid size N on the x-axis. Measure relative error as

$$Err = \frac{\|f'(x) - f'_n(x)\|_p}{\|f'(x)\|_p}.$$

Plot the error using both the $p = \infty, 2$ norms in a log-log scale. The relative errors should be lines with slopes 1 and $3/2$ respectively, i.e. it scales with $\frac{1}{N}, \frac{1}{N^{3/2}}$. Extra Credit: The $3/2$ slope is strange, why is that happening?

Result:

(a)

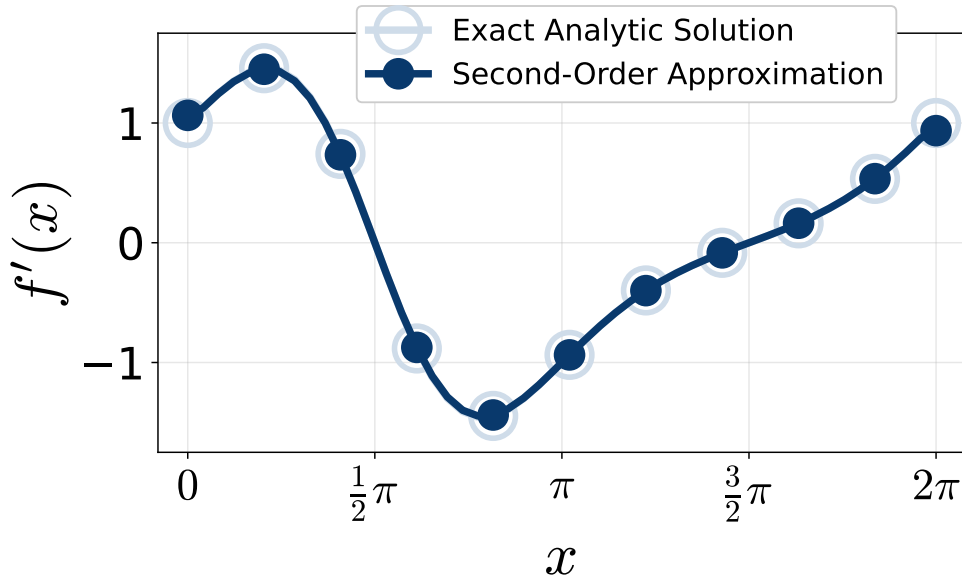


Figure 1.1: Accuracy of Second-Order approximation of the given function's derivative with $N = 50$.

(b)

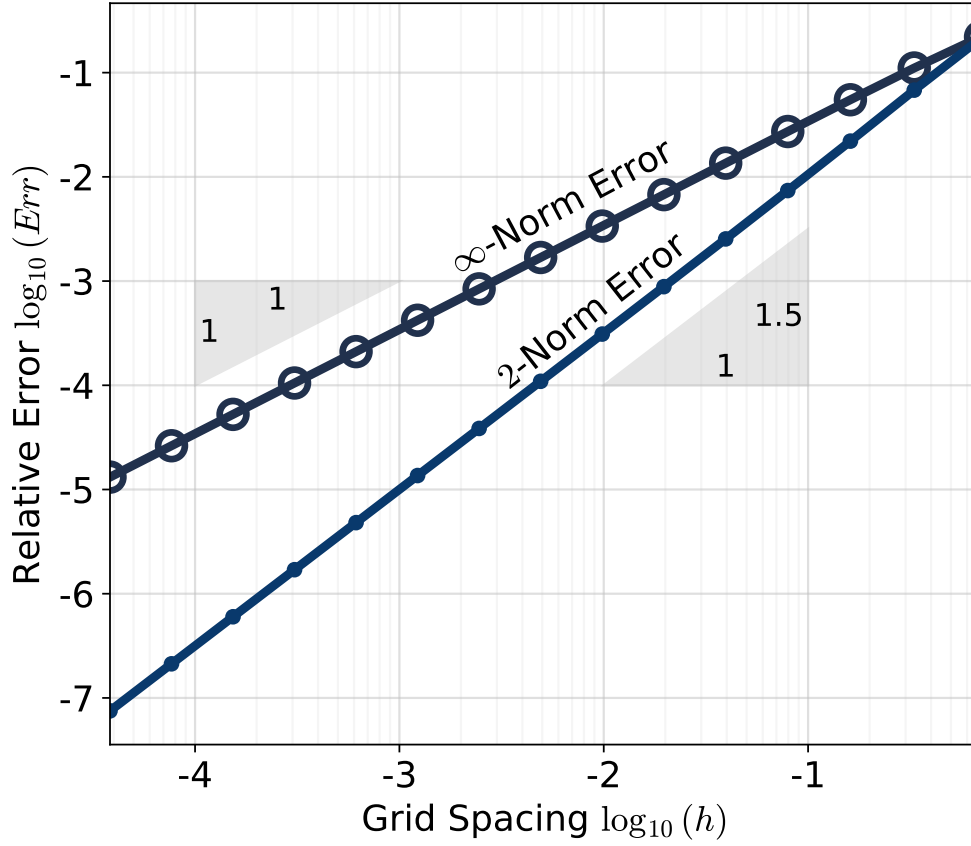


Figure 1.2: p -norm convergence for $p \in \{2, \infty\}$ of the derivative approximation.

Proof. We begin with the given equation for the 2-norm error:

$$Err_2 = \frac{\|f'(x) - f'_n(x)\|_2}{\|f'(x)\|_2}$$

When order notation is introduced, the $3/2$ slope property appears. To begin, state that for all x_i , its derivative does not depend on h :

$$f'(x_i) \rightarrow O(1), \quad i \in \{1, 2, \dots, N\}$$

Next, consider the error of the first-order-evaluated endpoints, for which it is well known their error scales with h :

$$|f'(x_i) - f'_n(x_i)| \rightarrow O(h), \quad i = 1, N$$

Similarly, the error for the second-order-evaluated interior points:

$$|f'(x_i) - f'_n(x_i)| \rightarrow O(h^2), \quad i \in \{2, 3, \dots, N-1\}$$

Therefore the 2-norm of the error at x_i and the analytic solution at x_i can be calculated:

$$\|f'(x) - f'_n(x)\|_2 = \left(\sum_{i=1}^N f'(x_i) - f'_n(x_i)^2 \right)^{1/2} \rightarrow \sqrt{O(h)^2 + O(N) \cdot O(h^2)^2},$$

where $O(h)$ comes from the two endpoints' error. Following simplifications:

$$\begin{aligned} O(h)^2 &= O(h^2), \\ O(N) \cdot O(h^2)^2 &= O(h^3) \quad (\text{by } N \rightarrow O(1/h)), \\ O(h^2) + O(h^3) &= O(h^2) \quad (\text{by } h \text{ small}), \\ \sqrt{O(h^2)} &= O(h) \end{aligned}$$

the order of the numerator term (that is, the norm of the error at every x_i) is yielded:

$$\|f'(x) - f'_n(x)\|_2 \rightarrow O(h)$$

A similar and more straightford process occurs for the denominator term (that is, the norm of the analytic solution at every x_i):

$$\|f'(x)\|_2 = \left(\sum_{i=1}^N f'(x_i) \right)^{1/2} \rightarrow \sqrt{O(N)} = O(h^{-1/2})$$

Therefore the order for the convergence emerges:

$$Err_2 \rightarrow \frac{O(h)}{O(h^{-1/2})} = O(h^{3/2}) = O\left(\frac{1}{N^{3/2}}\right)$$

□

Problem 2: Second-order derivatives Repeat problem 1 but compute the second derivative $f''(x)$. Assume periodicity and wrap around the grid instead of using one-sided finite differences.

Result:

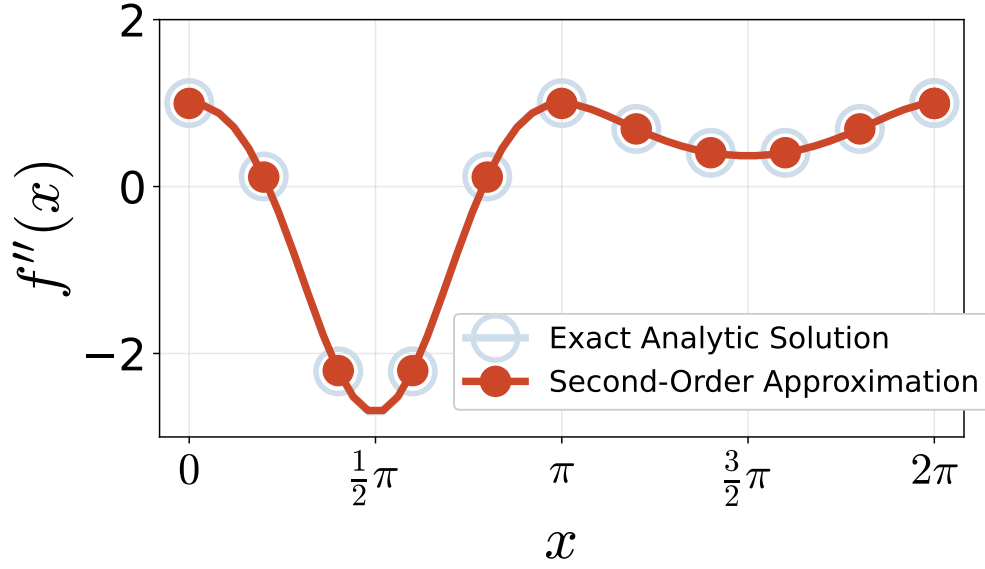


Figure 2.1: Accuracy of Second-Order approximation of the given function's second derivative with $N=50$.

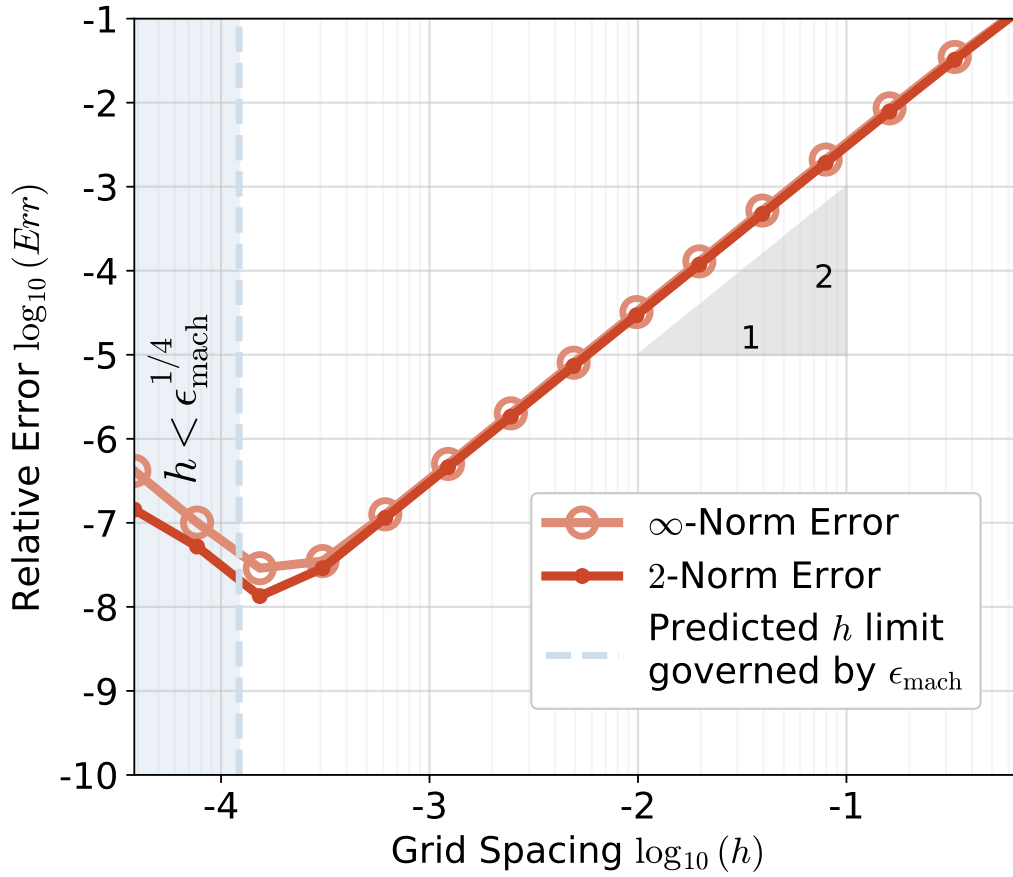
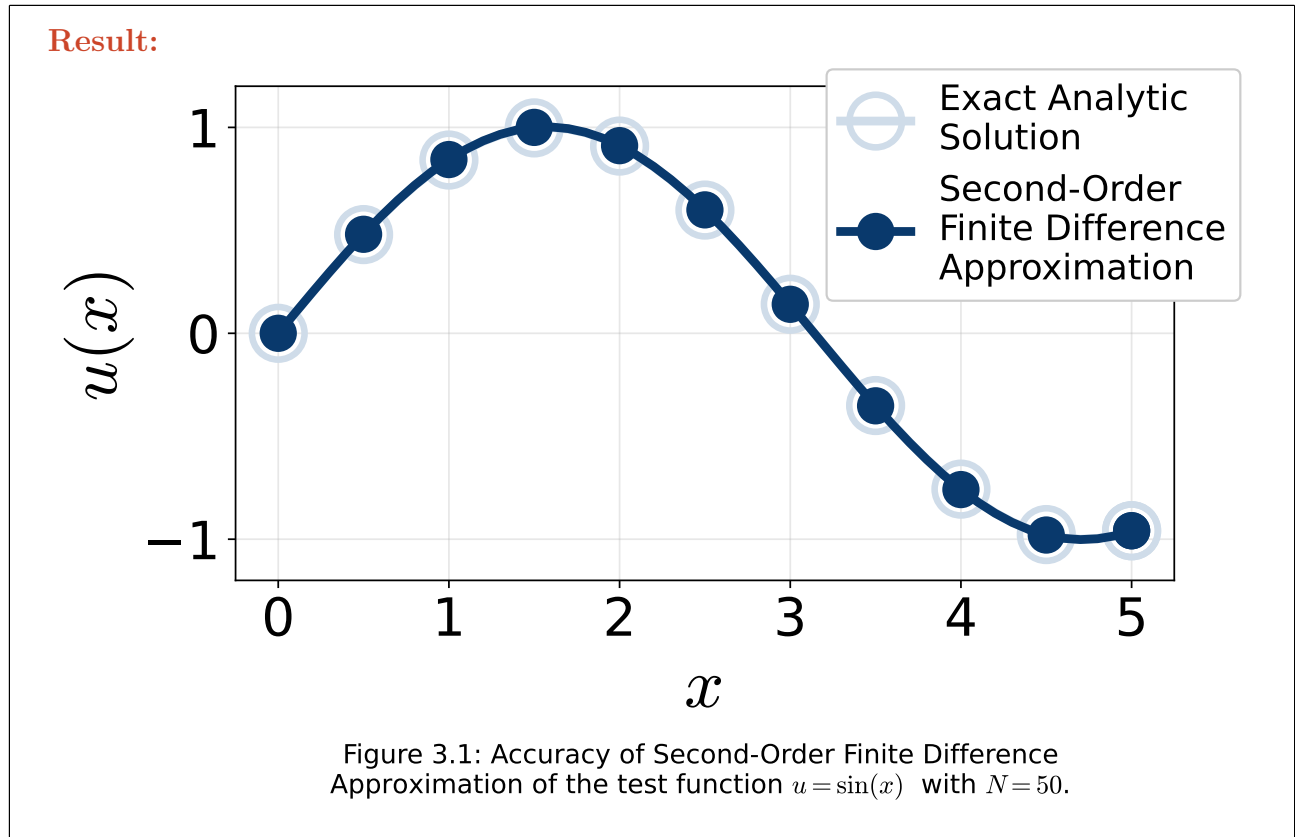


Figure 2.2: p -norm convergence for $p \in \{2, \infty\}$ of the second-derivative approximation.

Problem 3: ODE Consider the ODE

$$\frac{d^2u}{dx^2} + \sin(x) \frac{du}{dx} + u(x) = f(x)$$

defined on $0 \leq x \leq 5$. Solve the ODE using second-order finite difference methods with the Dirichlet conditions. Use $u(x) = \sin(x)$ to test your solution by finding boundary conditions and the necessary forcing function $f(x)$. Show an error plot that proves your method converges with second order.



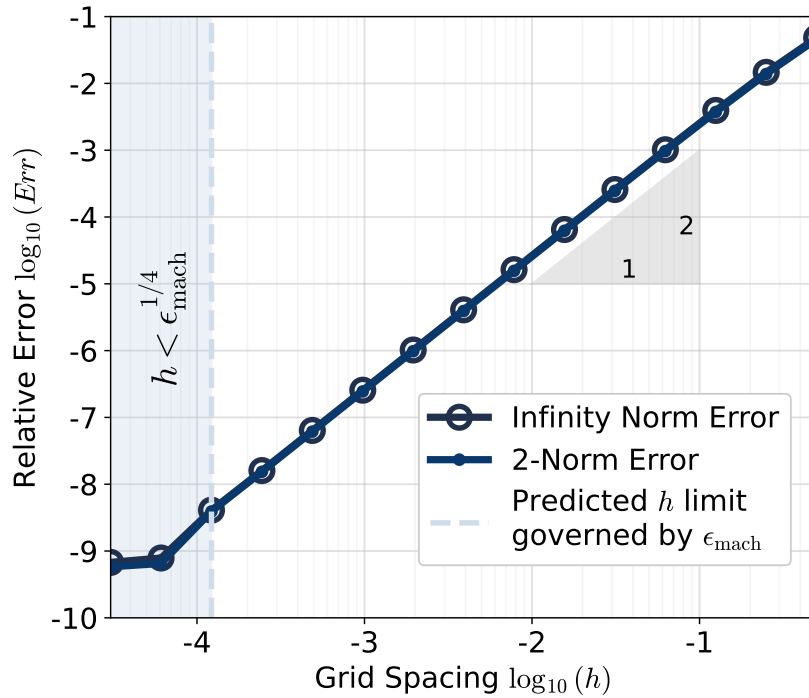


Figure 3.2: p -norm convergence for $p \in \{2, \infty\}$ of the finite difference approximation.

Problem 4: Poisson Equation Use finite difference methods to solve the Poisson equation $\nabla^2 u = f(x, y)$ in the domain $x \in [0, 5], y \in [0, 5]$. Use the test solution $u(x, y) = \sin(x) \cos(y)$.

- Apply Dirichlet conditions on all boundaries so that $u(x, y)$ satisfies the PDE and the boundary conditions, and show a plot of your solution and a second order convergence plot (similar to the first three problems).
- Modify your code to apply Neumann conditions on all boundaries. **This won't work.** Why not? – both in terms of the linear algebra and in terms of the actual PDE we want to solve. Suggest a solution to the issue.

Result:

(a)

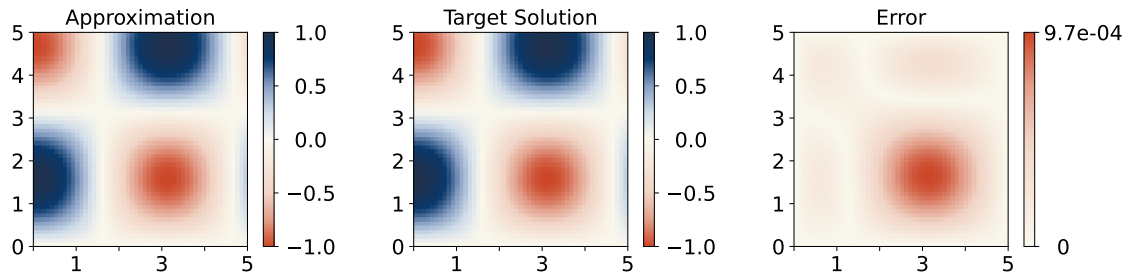


Figure 4.1: Accuracy of Second-Order Finite Difference Approximation of the test function $u = \sin(x)\cos(y)$ with $N = 50$, using known value for u at the boundaries.

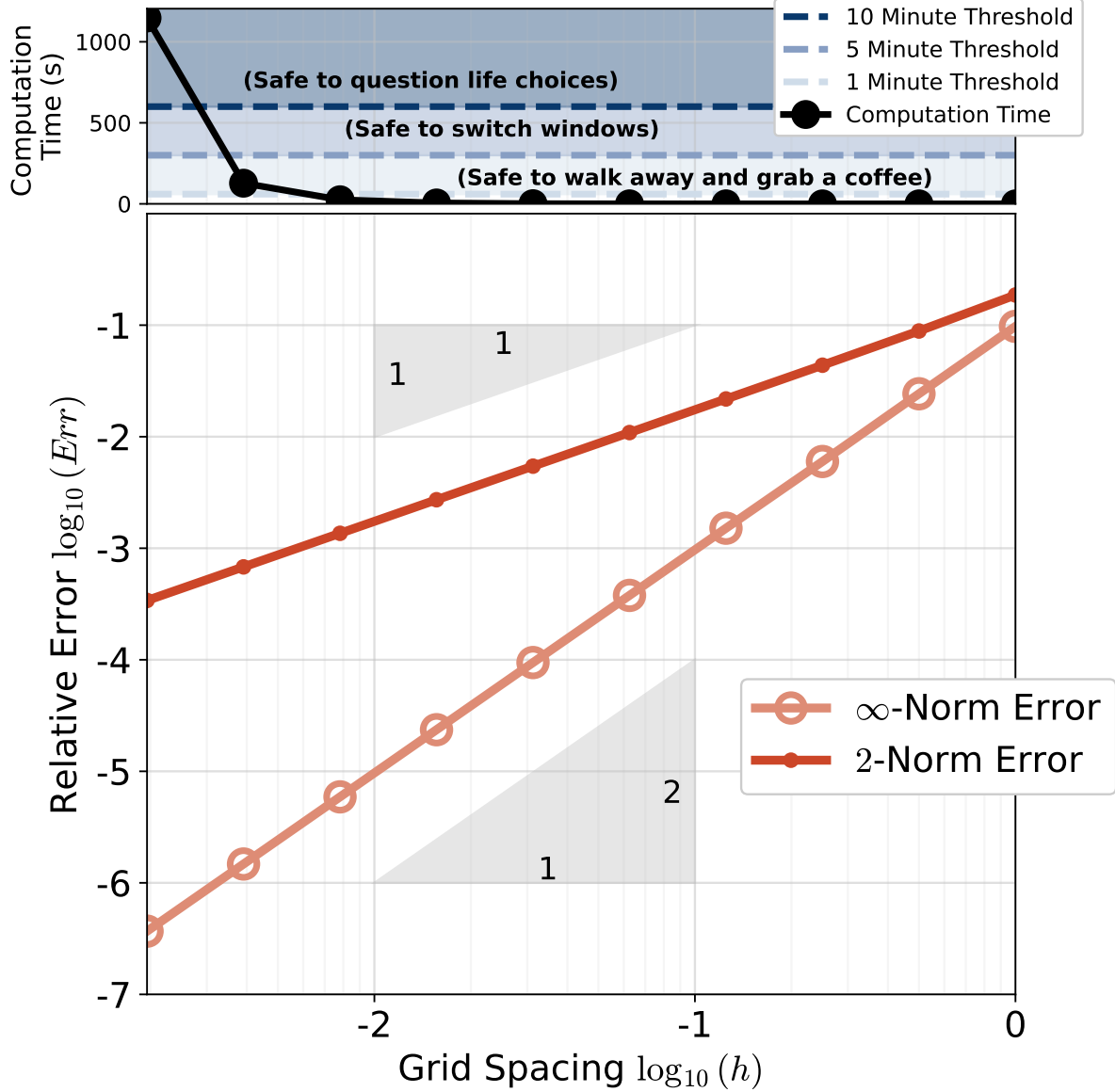


Figure 4.2: p -norm convergence for $p \in \{2, \infty\}$ of the Laplacian 5-point stencil approximation.

(b) *Proof.* Explanation of why Neumann fails:

The governing equations are only concerned with the first and second derivatives of u .

The fundamental theorem of calculus reminds us that integrating an undefined function gets us an arbitrary constant:

$$\int g(x) dx = \int g(x) dx + C$$

That tells us:

u solves some property based on $u_x, u_y \implies u + C$ solves the same property.

Therefore, the linear equation for $u_{i,j}$ is *underdetermined* and will be correct, $\pm$ a constant, C . Which, based on the solver used, can be close to the maximum magnitude the machine can compute. $\square$

Neumann can be accurate to give or take a constant of **2.3 100 billions**.

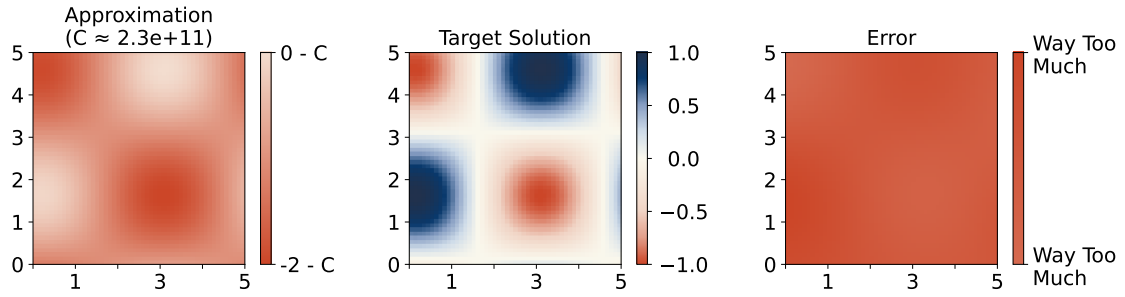


Figure 4.3: Accuracy of Second-Order Finite Difference Approximation of the test function $u = \sin(x)\cos(y)$ with $N = 50$. using known value for u_x, u_y at the boundaries.

However, let's say we know that this solution is correct to a constant, albeit a large one. If we also know what the minimum of the function *should* be, then we can adjust accordingly:

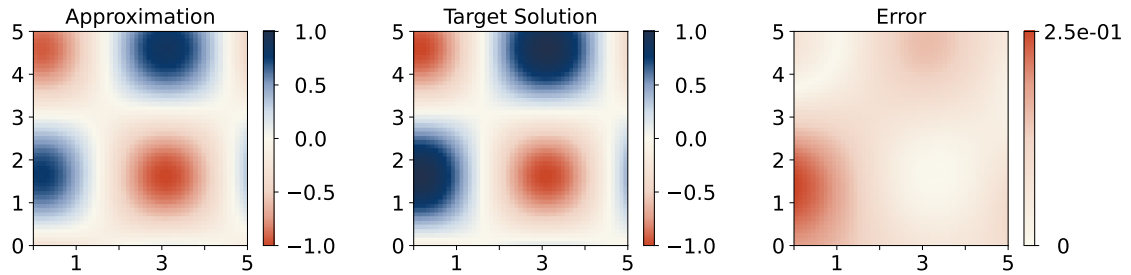


Figure 4.4: Accuracy of Second-Order Finite Difference Approximation of the test function $u = \sin(x)\cos(y)$ with $N = 50$. using known value for u_x, u_y at the boundaries and adjusting to a known minimum.

Alternatively, if we know a property like oscillation about zero, we can enforce that the boundary's norm is zero (that is, $u(x, y)$ sums to zero on the boundary):

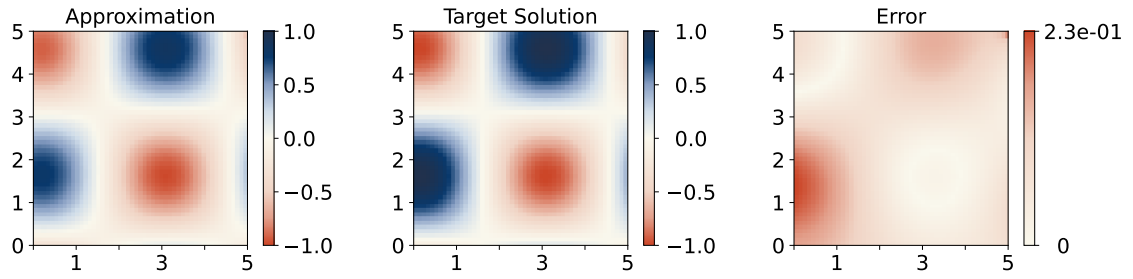


Figure 4.5: Accuracy of Second-Order Finite Difference Approximation of the test function $u = \sin(x)\cos(y)$ with $N = 50$, using known value for u_x, u_y at the boundaries and constraining the boundary norm to be zero.